Alexander Torres

Curriculum Vitae

torre280@msu.edu

Undergraduate Education – The Ohio State University: GPA 3.87

Aug 2022- May 2026 **B.Sc - Physics**, *GPA from Physics Courses - 3.98* Honors

B.Sc - Astronomy & Astrophysics, *GPA from Astronomy & Astrophysics Courses - 4.00* Honors

Minor - Mathematics

Minor - Spanish

Study Abroad – Universitat de València: GPA 4.00

Jun 2023 **Coursework**, *Panorama of Spanish Literature*, *Spanish Cinema*

Research Experience

Jan 2024 - Present **Research under Dr. Klaus Honscheid at The Ohio State University**, *Analyzing galaxy fields with machine learning methods to increase precision in constraints of the Alcock-Paczynski (AP) effect; using Bayesian statistics to characterize the time dependency of dark energy from the AP effect; identification of consistent projection effects in constraints of dark energy equation of state parameters from analysis of DESI baryonic acoustic oscillation distance data; novel measurement of Hubble constant to resolve tension in existing values from local distance ladder and cosmic microwave background measurements; guided study of cosmology fundamentals*

May 2025 - Aug 2025 **Research under Dr. Bangalore Sathyaprakash at Pennsylvania State University**, *Developed phenomenological, data-driven alternatives to the standard post-Newtonian approach for predicting time of common horizon formation in generic astrophysical binary black hole systems using techniques in symbolic regression and machine learning; received overview of gravitational waveform modeling and analysis; interpreted numerical relativity simulation data run using the SpEC code; self study of the quasi-local horizon framework for characterizing black holes*

Feb 2023 - May 2023 **Research under Dr. Nandini Trivedi at The Ohio State University**, *Received an overview of quantum information and basic quantum computing; obtained experience in quantum computing logic gates with the application qBraid; guided study of quantum mechanics fundamentals*

Honors

2026 **Graduate Research Fellow**, *National Science Foundation*

2026 **Poster Session Top Presenter**, *American Physical Society Global Summit*

2026 **Graduated Magna Cum Laude**, *The Ohio State University*

2023 - 2026 **Dean's List**, *The Ohio State University*

2022 **Recipient of Valentino Physics Academic Achievement Scholarship**, *The Ohio State University Department of Physics*

Talks

Jul 2025 **Common Horizon Formation in Simulations of Binary Black Hole Mergers**, *Torres A, Sathyaprakash B, and Prasad V*, Presented at Penn State REU Symposium, Pennsylvania State University

Jul 2025 **Crash Course in Gravitational Waves**, *Torres A, Rocha A*, Presented at AstroFest Outreach Program, Pennsylvania State University

Poster Presentations

Mar 2026 **Common Horizon Formation in Simulations of Binary Black Hole Mergers**, *Torres A, Sathyaprakash B, and Prasad V*, Presented at the APS Global Physics Summit, Denver

Jul 2025 **Common Horizon Formation in Simulations of Binary Black Hole Mergers**, *Torres A, Sathyaprakash B, and Prasad V*, Presented at Penn State REU Symposium, Pennsylvania State University

Jul 2024 **Probing Potential Projection Effects in DESI Constraints of Dark Energy Equation of State Parameters**, *Torres A, Honscheid K, Taylor P, and Zaborowski E*, Presented at Department of Physics SRP Poster Session, The Ohio State University

Apr 2023 **Classical Scattering Models with Coulomb and Nuclear Potentials**, *Torres A and Sharma P*, Presented at Polaris Undergraduate Poster Show, The Ohio State University

Publications

Jun 2025 **A Sound Horizon-Free Measurement of H0 in DESI 2024**, *E.A. Zaborowski et al.*, As a co-author, I tested the novel analysis method presented in this paper on mock data from AbacusSummit simulations (Fig. 2)

Leadership and Mentoring Experience

August 2025 - Present **Physics Tutor**, *OSU Department of Physics*, Guide students with widely varying prior physics experience through introductory mechanics and electricity & magnetism problems; teach concepts and problem solving techniques while reviewing specific examples relevant to the intro courses

Mar 2024 - May 2025 **President**, *OSU Sigma Pi Sigma Chapter (Physics and Astronomy Honor Society)*, Organized professional development activities and research talks to provide extracurricular enrichment for undergraduates; established annual trip to Fermilab to expose students to cutting-edge physics research facilities; attended American Physical Society conferences affiliated with the national Sigma Pi Sigma organization; oversaw members and managed responsibilities of chapter executive board; served as a liaison between the chapter and university; communicated with the national Sigma Pi Sigma organization

Mar 2023 - May 2024 **Secretary**, *OSU Astronomical Society*, Sent out regular club communications; advertised enriching department and campus opportunities; managed monthly club newsletter; wrote popular science articles for club website; networked with pertinent physics/astronomy department professionals and other university entities on the club's behalf; organized and volunteered at community stargazing and astronomy outreach events

Jan 2023 - Jul 2023 **Assembly Fellow**, *Access Network and Polaris*, The Access Network develops and advances policies and practices in higher education that nurture supportive communities in which students are centered and exercise agency in their education; I organized annual Access Network Assembly logistics such as housing, food, and communications; I prepared and shared presentations during the assembly over topics such as the implementation of programming as a tool in STEM as well as how to increase student confidence in programming abilities

Course Projects

Feb 2023 - Apr 2023 **Classical Scattering Models with Coulomb and Nuclear Potentials**, *Investigation of differential cross sections, numerically solving differential equations, relative small-scale behavior of the Coulomb and Woods-Saxon nuclear potential*

Relevant Skills

Programming, *Items marked "Fluent" indicate sufficient comfort to implement projects with little to no reliance on language specific reference. Items marked "Familiar" indicate comfort with reading the language and doing projects with language specific references.*

- Python, Fluent
- Mathematica, Familiar

Other Computational Skills

- Comfortable with machine learning via PyTorch
- Comfortable coding on command line, using Vim, and working remotely via SSH
- Comfortable with LaTeX